

A Coherent-Hilbert Subcase of the UMD Openness Conjecture and Its Endpoint Bilinear-Hilbert Consequence

Candidate partial result for arXiv:1909.06416

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Status. Candidate partial; likely valid, subject to human review. The argument proves Conjecture 8.9 for a coherent two-stage interpolation subclass. It does *not* synchronize arbitrary Hilbert endpoints and does not claim the full conjecture.

1 Source question

The source is A. Amenta and G. Uraltsev, *The bilinear Hilbert transform in UMD spaces* [1]. The target is Conjecture 8.9 and Remark 8.10 on PDF page 67 of arXiv:1909.06416. For $r \in [2, \infty)$, a Banach space X is called r -intermediate UMD when there are a compatible couple (Y, H) , with Y UMD and H Hilbert, such that

$$X \simeq [Y, H]_{2/r}$$

for the complex interpolation functor.

However, when linked with $\Pi = \text{BHF}$, the triple could hypothetically be thought of as a non-UMD triple, even though the spaces L^{p_i} are themselves UMD.

The only remaining question is what happens when

$$(8.9) \quad \sum_{i=1}^3 \frac{1}{r_i} = 1,$$

which sits between of our main result and our counterexample. We conjecture that Theorem 1.1 holds under this endpoint assumption. In fact, we will see that it follows from an openness property of intermediate UMD spaces that we also conjecture.

Conjecture 8.9 (UMD openness conjecture). Let X be an r -intermediate UMD space for some $r \in (2, \infty)$. Then there exists $\tilde{r} < r$ such that X is $\tilde{r}$ -intermediate UMD. Equivalently, the set of exponents $r \in [2, \infty)$ such that X is r -intermediate UMD is relatively open in $[2, \infty)$.

This can be thought of as an extension of the intermediate UMD conjecture, which says that every ∞ -intermediate UMD space (that is, every UMD space) is r -intermediate for some $r < \infty$.

Assuming the UMD openness conjecture, the intermediacy condition in Theorem 1.1 can be simply relaxed to (8.9). To see this, assuming (8.9), note that there is an index $j \in \{1, 2, 3\}$ such that $r_j > 2$. By the UMD openness conjecture, there exists $\tilde{r}_j \in [2, r_j)$ such that X_j is $\tilde{r}_j$ -intermediate UMD. Now we have

$$\frac{1}{\tilde{r}_j} + \sum_{i \neq j} \frac{1}{r_i} > 1,$$

Figure 1: Full-width raster crop of source PDF page 67, containing the endpoint question, the complete statement of Conjecture 8.9, and its immediate BHT consequence.

Transcription. Conjecture 8.9 asks: if X is r -intermediate UMD for some $r \in (2, \infty)$, must there exist $\tilde{r} < r$ such that X is $\tilde{r}$ -intermediate UMD? The source observes that a positive answer would treat its remaining endpoint regime

$$\sum_{i=1}^3 \frac{1}{r_i} = 1. \quad (1)$$

Remark 8.10 sketches a reiteration proof but correctly warns that the Hilbert spaces in two successive representations need not be the same compatible endpoint.

2 Idea of the proof

The source's warning identifies the whole issue. If the two stages really do use one Hilbert endpoint H , they are not merely abstractly similar: they lie on one complex interpolation scale. Reiteration then combines their two positive Hilbert weights. The combined interpolation parameter is strictly larger than $2/r$, hence its corresponding intermediate exponent is strictly smaller than r .

For an endpoint BHT triple, improving a single $r_j > 2$ makes the reciprocal sum in (1) strictly greater than one. This opens a genuine region of Hölder exponents to which the source's Theorem 1.1 applies.

3 The coherent-Hilbert theorem

Definition 1. Let $r > 2$ and put $\theta = 2/r$. An r -intermediate representation of X is *coherently twice intermediate* if there are one compatible couple (Z, H) , where Z is UMD and H is Hilbert, and an $\alpha \in (0, 1)$ such that

$$X \simeq [[Z, H]_\alpha, H]_\theta. \quad (2)$$

Thus the left endpoint in the outer representation is itself obtained by moving toward the *same* Hilbert endpoint.

Theorem 1 (strict improvement for a coherent endpoint). *Suppose (2) holds. Set*

$$\beta = \theta + \alpha(1 - \theta), \quad \tilde{r} = \frac{2}{\beta}. \quad (3)$$

Then

$$2 < \tilde{r} < r \quad \text{and} \quad X \simeq [Z, H]_{2/\tilde{r}}.$$

In particular, X satisfies the conclusion of the UMD openness conjecture at r .

Proof. The complex reiteration theorem, in its endpoint form, gives

$$[[Z, H]_\alpha, H]_\theta = [Z, H]_{(1-\theta)\alpha+\theta} = [Z, H]_\beta;$$

see Bergh–Löfström, Theorem 4.6.1 [2], which is also the theorem invoked in source Remark 8.10. Since $0 < \alpha < 1$ and $0 < \theta < 1$,

$$\theta < \theta + \alpha(1 - \theta) < 1.$$

Consequently $2 < 2/\beta < 2/\theta = r$. Since Z is UMD and H is Hilbert, the last interpolation identity is exactly a $\tilde{r}$ -intermediate UMD representation. $\square$

For context, the easy direction of the exponent geometry is automatic.

Proposition 1 (upward-ray property). *If X is r_0 -intermediate UMD, then it is r -intermediate UMD for every $r > r_0$.*

Proof. Write $X \simeq [Y, H]_{\theta_0}$ with $\theta_0 = 2/r_0$, and let $\theta = 2/r < \theta_0$. Put

$$\delta = \frac{\theta_0 - \theta}{1 - \theta} \in (0, 1), \quad W = [Y, H]_{\delta}.$$

The space W is UMD because the UMD property is stable under complex interpolation. Reiteration gives $[W, H]_{\theta} = [Y, H]_{(1-\theta)\delta+\theta} = [Y, H]_{\theta_0} \simeq X$. $\square$

This proposition shows that Conjecture 8.9 concerns whether the left endpoint of this upward ray is ever attained above 2. The coherent theorem excludes attainment for the stated subclass.

4 Endpoint bilinear Hilbert transform consequence

We use the notation of source Theorem 1.1. Let X_i be r_i -intermediate UMD, let $\Pi : X_1 \times X_2 \times X_3 \rightarrow \mathbb{C}$ be a bounded trilinear form, and let BHF_{Π} denote the associated trilinear bilinear-Hilbert form.

Corollary 1 (nonvacuous endpoint subcase). *Assume (1). Suppose some index j with $r_j > 2$ has a coherently twice intermediate representation, and let $\tilde{r}_j$ be given by (3). Define*

$$s_j = \tilde{r}_j, \quad s_i = r_i \quad (i \neq j).$$

Then $\sum_i s_i^{-1} > 1$, and source Theorem 1.1 gives

$$|\text{BHF}_{\Pi}(f_1, f_2, f_3)| \lesssim \prod_{i=1}^3 \|f_i\|_{L^{p_i}(\mathbb{R}; X_i)}$$

for every Hölder triple $p_i \in (1, \infty)$ satisfying

$$\sum_{i=1}^3 \frac{1}{\min(p_i, s_i)'(s_i - 1)} > 1. \tag{4}$$

Moreover, the set of such Hölder triples is nonempty.

Proof. Only the j th exponent changes, so

$$\sum_i \frac{1}{s_i} = \frac{1}{\tilde{r}_j} + \sum_{i \neq j} \frac{1}{r_i} > \sum_i \frac{1}{r_i} = 1.$$

The theorem above makes X_j s_j -intermediate, while the other spaces retain their original representations. Source Theorem 1.1 therefore proves the asserted bound whenever (4) holds.

To see that this condition is not vacuous, choose numbers $a_i \in (0, 1/s_i]$ with $\sum_i a_i = 1$; such a choice exists because $\sum_i 1/s_i > 1$. Put $p_i = 1/a_i$. Then (p_i) is a finite Hölder triple and $p_i \geq s_i$, so

$$\frac{1}{\min(p_i, s_i)'(s_i - 1)} = \frac{1}{s_i'(s_i - 1)} = \frac{1}{s_i}.$$

Summation gives (4). $\square$

5 Stress tests and limitations

- The strict inequalities in (3) use both $r > 2$ and $0 < \alpha < 1$; no endpoint convention is hidden.
- The endpoint BHT corollary uses the improved tuple (s_i) in the source exponent condition. With the original endpoint tuple, every summand in that condition is at most $1/r_i$, so its admissible region would be empty.
- Scalar L^p and Schatten interpolation scales provide consistency checks because their standard complex interpolation identities use one fixed L^2 or S^2 endpoint. Bounds for those classical examples are already known and are not claimed as novel here.
- The proof does not turn two abstract representations $X \simeq [Y_1, H_1]_{2/r}$ and $Y_1 \simeq [Y_2, H_2]_{2/s}$ into a coherent one. An isomorphism $H_1 \simeq H_2$ need not extend to an isomorphism of compatible couples. That is the unresolved core of Conjecture 8.9.
- Eight focused upgrade attempts were made: arbitrary endpoint isomorphisms, Hilbert stabilization, graph/pushout constructions, subquotient interpolation, type/cotype inversion, lattice factorization, endpoint limiting, and reversal of the tensor-BHT counterexample. Each either reproduces the compatibility gap or fails an exact hypothesis; details are in the accompanying attempt note.

6 Novelty and review recommendation

Run indexes and bounded searches by arXiv id, exact conjecture name, source title, and the phrases “UMD openness” and “intermediate UMD” found no later claimed solution. This packet should nevertheless be viewed as a rigorous structural completion of the source’s own Remark 8.10, not as a claim that the full conjecture has been solved. Novelty confidence is therefore low; the mathematical value is the precise sufficient hypothesis and the explicit nonvacuous endpoint-BHT corollary.

Human review should check (i) that the intended compatible-couple identifications in (2) are exact enough for endpoint reiteration, and (ii) that the exponent region in the corollary is stated with the improved tuple. Once those points are accepted, the proof is a direct application of the standard reiteration theorem.

References

- [1] A. Amenta and G. Uraltsev, *The bilinear Hilbert transform in UMD spaces*, Math. Ann. 378 (2020), 1129–1221; arXiv:1909.06416.
- [2] J. Bergh and J. Löfström, *Interpolation Spaces: An Introduction*, Grundlehren der mathematischen Wissenschaften 223, Springer, 1976.